[INCH-POUND]
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22 August 2024

COMMERCIAL ITEM DESCRIPTION

MILLING MACHINE, VERTICAL, RAM AND TURRET, KNEE, MANUAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers manual, knee, ram and turret, vertical milling machines. The machines are intended for shaping workpieces of cast iron, steel, titanium, aluminum, brass, graphite epoxy, and composite materials by the removal processes of drilling, reaming, tapping, milling, and boring for use in Aircraft Metals Technology Shops across the Air Force. The equipment identified herein should be manufactured in the United States of America.

2. CLASSIFICATION. The milling machine shall be of the following characteristics specified in Table I.

Table I. Milling machine characteristics.

Table Size	X-Axis Travel	Y-Axis Travel	Z-Axis Travel
9 in. x 49 in.	34-36.6 in.	12-14.2 in.	15-17.2 in.

Note 1: Offered sizes shall not be less than, nor exceed 10 percent greater than, the requirements stated in Table I.

3. SALIENT CHARACTERISTICS.

3.1 General requirements. The milling machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. The milling machine shall be constructed from materials with sufficient mass and strength and configured in a shape and size that shall support the workpiece and machine components while minimizing element deflections and initiated and/or transmitted vibration during operation. The rotation of the axis of the spindle shall be vertical. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

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3.1.1 Identification of machine axes. The X-, Y-, and Z-axes of movement of the milling machine shall be identified in accordance with Electronic Industries Association (EIA) standard 267.

3.2 Components. Access doors or removable panels shall be provided for servicing all components located within. The milling machine shall include, as a minimum, the following components:

3.2.1 Column and base. The column and base shall support the other elements of the milling machine and shall incorporate materials, shape, and mass to minimize adverse vibration. The base shall have provisions for leveling and anchoring the machine.

3.2.2 Base anchoring. The milling machine shall be delivered with fasteners, adapters, clamps, or other hardware required to affix the base to the floor. Maximum diameter for each hole required in the shop floor shall be 1 inch.

3.2.3 Knee. The knee shall be fitted to the column machine ways and shall have machine ways to support the saddle. The knee shall have vertical (Y-axis) traverse by means of an elevating screw operated by a crank or handwheel and power feed.

3.2.4 Saddle. The saddle shall have machine ways to support the worktable. The saddle shall have transverse (Z-axis) traverse by means of a screw operated by a crank or handwheel and power feed.

3.2.5 Machine ways. The machine ways shall be the standard design of the manufacturer. A means shall be provided to prevent dust, cutting fluids, and chip swarf from accumulating on machine ways and becoming lodged between sliding parts. Machine ways shall be machined/ground smooth to allow continuous, uninterrupted movement of the knee, guides, and slides for machining operations. Machine ways shall not corrode or degrade under normal operations and maintenance conditions and/or storage for no less than fifteen (15) years from the date of procurement.

3.2.6 Worktable. The worktable length, width, and weight-holding capacity shall be provided as specified in Table I and considered by the manufacturer as standard. The worktable shall have a quantity of (3) T-slots of the size and spacing considered by the manufacturer as standard. The dimensions of the T-slots shall conform to ANSI B5.1M. The worktable shall have longitudinal (X-axis) traverse by means of a screw operated by a crank or handwheel and power feed.

3.2.7 Spindle head assembly. The spindle quill shall have power feed at three rates as specified in Table II, adjustable automatic feed disengagement in both directions, and a micrometer depth stop mechanism. The spindle motor shall provide variable spindle speeds in accordance with the manufacturer's standard machine design and the ranges specified in Table II. The spindle taper shall be of either the NMTB-30 or size 30 Quick Change varieties. The spindle head assembly shall swivel 90 degrees right and left from vertical and 45 degrees front and back from vertical. The spindle head assembly shall include a spindle hand brake capable of stopping rotation of the spindle. A power drawbar compatible with the spindle head assembly shall be furnished.

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Table II: Spindle requirements.

Low Speed Range	High Speed Range	Quill Feed Rates	Quill Travel	Spindle Motor Rating
Min: 50-65 rpm Max: 400-525 rpm	Min: 400-550 rpm Max: 4200-4500 rpm	0.0015 in./rev. 0.003 in./rev. 0.006 in./rev.	5-5.25 in.	3 hp

Note 2: Spindle speeds and motor ratings shall not be less than, nor exceed 10 percent greater than, the requirements detailed in Table II.

3.2.8 Ram. The milling machine shall have an overarm ram which moves the spindle assembly a minimum of 12 inches horizontally across the worktable. Ram positioning shall be by manual means.

3.2.9 Turret. The milling machine shall have a turret that allows the ram and spindle assembly to rotate 360 degrees around the vertical axis of the column, or neck, by manual means.

3.2.10 Locking devices. The milling machine shall be equipped with locking or clamping devices to hold the ram, turret, spindle assembly, table, saddle, and knee at any position in their range of operation.

3.2.11 Dials and indicators. Angular scales shall be divided into 1-degree, or smaller, increments. Linear scales shall be divided into 0.001-inch or 0.02-millimeter increments. The zero-degree position of angular scales shall be prominently marked.

3.2.12 Controls. The milling machine controls shall include, as a minimum, the following functions:

a. Power on/off.

b. Spindle start/stop.

c. Spindle-speed.

d. Spindle hand brake.

e. X- and Y-axis motion (power feed and manual).

f. Z-axis motion.

g. Turret rotation.

h. Ram motion.

i. Spindle assembly motion.

3.2.13 Digital readout system. The milling machine shall be equipped with a digital readout system consisting of sensors and display for the X-, Y-, and Z-axes of travel. The digital readout system shall have a minimum resolution of 0.0002 inches.

3.2.14 Lubrication system. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. All oil holes, grease fittings, and filler caps shall be accessible, and a manually operated “one shot” lubrication system shall be provided to deliver oil to the table,

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saddle, and knee machine ways and leadscrews. A sight gauge, warning light, or similar device shall be provided for determining lubricant level in the reservoir(s).

3.2.15 Coolant system. The milling machine shall have a mist coolant system. A means shall be provided to direct and control the amount of coolant to the cutting tool over the entire work area. A base- or floor-mounted splash tray shall be provided to catch coolant overflow.

3.2.16 Electrical system. The electrical system shall conform to the National Fire Protection Association (NFPA) 79 standard. The system shall allow for either 230 VAC three phase or 460 VAC three phase connections. The control system shall operate on a separate 120-volt, single-phase circuit from the machine main power supply.

3.2.17 Electric motors. The spindle-drive motor and motion-control motors shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG 1 and NEMA ICS 16 standards, respectively. The spindle-drive motor shall have an open drip-proof enclosure. Motion- and position-control motors and controls shall conform to NEMA ICS 1 and NEMA ICS 3.

3.2.18 Tooling, attachments, and accessories. All tooling, attachments, accessories, and hardware required to render the machine usable shall be included, including, as a minimum:

3.2.19 Endmill. A minimum of one indexable, ½” cut diameter, 5/8” shank, square-shoulder end mill shall be provided and shall be functional with the spindle head assembly. The endmill shall conform to ISO 13399 standards.

3.2.19.1 Tool package. A tool package appropriate for the general finishing of steel and cast iron shall be included with the milling machine. Tooling shall be compatible with the endmill provided.

3.2.19.2 Workholding. A removeable, manually operated vise shall be provided. The vise jaws shall be 6 inches in width, with a maximum opening size of 9 inches. Sine keys compatible with the worktable T-slots shall be provided to securely mount the vise to the worktable.

3.2.19.3 Work light. An adjustable position work light compatible with the milling machine shall be provided. The work light shall be of adequate brightness to illuminate the worktable.

3.2.19.4 Manuals. Two sets of the milling machine installation, operation, and maintenance manuals shall be provided.

3.3 Warranty. A minimum one-year warranty shall be provided to protect against defects, breaks, or component failures during normal operations. The warranty shall include the replacement and installation of components/assemblies at the installation location by an OEM-certified technician. The warranty does not cover tooling identified in 3.2.19.1 and 3.2.19.2. For equipment located in austere operating environments, an OEM-certified technician may oversee the repair by a Government representative; however, replacement components shall still be provided as part of the warranty.

3.4 Support. Technical support for the equipment shall be provided for a period of three years after the date of purchase. Technical support shall be accessible, at a minimum, via phone and email.

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3.5 Performance. The milling machine shall meet the performance requirements of sections 5.3.1 through 5.3.3.

3.6 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.8. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall comply with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI B11.8 and OSHA standards, the OSHA standards shall apply.

3.7 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. All measuring and indicating devices on the machine shall be graduated in the same system.

3.8 Nameplate. A nameplate shall be securely attached to the machine and shall contain the following information:

a. Nomenclature.

b. Manufacturer's name.

c. Serial number.

d. Machine model designation.

e. Power input (volts, total amperes, phase, and frequency).

f. Short-circuit/over-current rating.

g. Contract number or order number.

h. National stock number.

i. Date of manufacture.

3.9 Lubrication plate or chart. A lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.

b. Servicing interval.

c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE.

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5.1 Product conformance. The products provided shall meet the salient characteristics (section 3) of this commercial item description; shall conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and shall be the same system offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The milling machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. The milling machine shall be tested in accordance with sections 5.3.1 through 5.3.3. Should a malfunction or a failure occur during any test, the problem shall be corrected, and the test shall be rerun from the beginning.

5.3.1 Test setup and conditions. The manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to the National Institute of Standards and Technology (NIST) and shall provide certification of calibration. The machine shall be tested in the environmental conditions considered by the manufacturer as standard for machine performance verification.

5.3.2 Operational test. The milling machine shall be operated in accordance with the manufacturer's standard operating procedures. The milling machine shall be operated for not less than four hours in which proper operation of all controls and features shall be verified.

5.3.3 Performance tests. The performance tests shall be conducted in accordance with procedures considered by the manufacturer as standard for machine performance verification.

5.3.3.1 Alignment tests. The milling machine shall meet the alignment specifications considered standard by the manufacturer.

5.3.3.2 Horsepower, milling accuracy, and tool chatter tests. The milling machine shall be tested by machining a workpiece of SAE 4140 steel. The workpiece shall not be less than 16 inches long, 6 inches wide, and 3 inches thick and may be drilled to facilitate clamping. One large surface shall be ground flat and mounted in contact with the table. Roughing cuts shall be made on the horizontal surface and on two adjacent vertical surfaces of the workpiece at sufficient depth and feed rate to load the spindle drive motor to at least the maximum duty rated electrical input. The type of cutting tool, depth of cut, spindle speed, and feed rate shall be determined by the manufacturer, provided the required loading is achieved. The resultant metal removal rate shall be at least 0.65 cubic inches per minute rated horsepower of the spindle drive motor. There shall be no evidence of tool chatter, motor stalling, or permanent distortion of the workpiece. Finishing cuts shall be made on the horizontal and vertical surfaces and shall meet the accuracy stated in 5.3.3.35 through 5.3.3.5.

5.3.3.3 Squareness. The two milled sides of the workpiece shall be measured for squareness. The sides shall be square with each other within 0.001 inch per foot.

5.3.3.4 Flatness. The two milled sides of the workpiece shall be measured for flatness. The sides shall be flat within 0.001 inch per foot.

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5.3.3.5 Parallelism. The finished horizontal surface of the workpiece shall be measured for parallelism. The top surface shall be parallel to the bottom surface within 0.001 inch per foot.

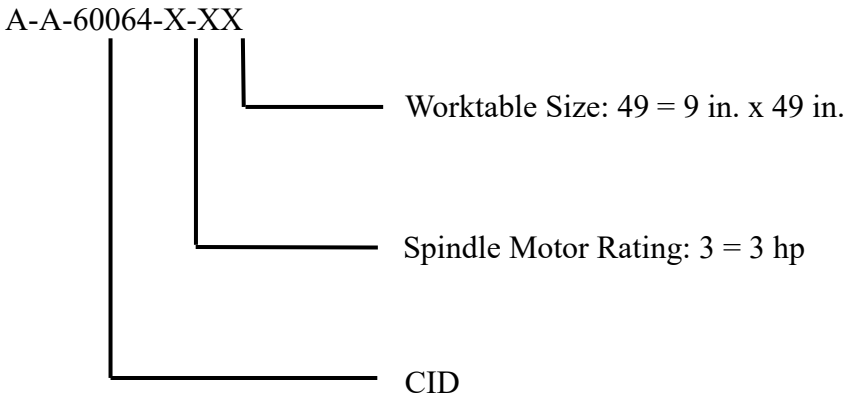
5.4 Acceptance. Preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared with the performance requirements described in sections 5.3.1 through 5.3.3, in addition to the performance characteristics considered standard by the manufacturer. Failure of the milling machine to meet any of the performance requirements shall be cause for rejection.

6. PACKAGING.

6.1 Preservation, packing, and marking. The products shall be preserved, packed, and marked as specified in the contract or purchase order.

7. NOTES.

7.1 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of referenced documents.

7.2.1 American National Standards Institute (ANSI). Copies of ANSI documents may be obtained at https://www.ansi.org/.

7.2.2 Electronic Industries Association (EIA). SAE International (formerly named the Society of Automotive Engineers). Copies of SAE documents may be obtained at: https://www.sae.org/.

7.2.3 Environmental Protection Agency (EPA). Copies of EPA documents may be obtained at https://www.epa.gov/.

7.2.4 National Electrical Manufacturers Association (NEMA). Copies of NEMA standards may be obtained at https://www.nema.org/.

7.2.5 National Fire Protection Association (NFPA). NFPA documents may be obtained at https://www.nfpa.org/.

7.2.6 International Organization for Standardization (ISO). Copies of ISO Standards may be obtained at https://www.iso.org.

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7.2.7 Occupational Safety and Health Administration (OSHA). Copies of OSHA documents may be obtained at https://www.osha.gov.

7.2.8 National Institute of Standards and Technology (NIST). Copies of NIST documents may be obtained at https://www.nist.gov/.

7.2.9 Federal Acquisition Regulation FAR. Electronic copies of the FAR may be obtained from https://www.acquisition.gov/far/index.html.

7.2.10 The Code of Federal Regulations (CFR) CFR documents. may be obtained at https://www.ecfr.gov/.

7.3 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.

b. Worktable size and spindle motor rating of milling machine required (see section 2).

c. Preservation, packing, and marking requirements.

7.4 Key words.

Boring

Drilling

Reaming

Tapping

Milling

MILITARY INTERESTS

Custodians:
Air force – 184

Reviewers:
GSA – FAS

Preparing Activity:
Air Force – 184

Agent:
AF – 110

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